CENTER FOR DRUG EVALUATION AND RESEARCH

APPLICATION NUMBER:

208400Orig1s000
208400Orig2s000

<u>CLINICAL PHARMACOLOGY AND BIOPHARMACEUTICS REVIEW(S)</u>

CLINICAL PHARMACOLOGY MEMORANDUM

Application Number: 208400, Original-2

Submission Date: October 25, 2016

Submission Type: 505(b)(2); Resubmission

Product: XATMEP (Methotrexate Oral Solution)

Proposed Indication: Polyarticular juvenile idiopathic arthritis (pJIA)

Applicant: Silvergate Pharmaceuticals, Inc.

OCP Division: Division of Clinical Pharmacology 2

Primary Reviewer: Manuela L T Grimstein, M.Sc., Ph.D.

Team Leader: Anshu Marathe, Ph.D.

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1. EXECUTIVE SUMMARY

Silvergate Pharmaceuticals, Inc. submitted NDA 208400 for Methotrexate Oral Solution, 2.5 mg/mL (XATMEP) as 505(b)(2) application on August 31, 2015. The application provides for a new oral solution dosage form for methotrexate, and is limited to the indications of acute lymphoblastic leukemia (ALL) and active polyarticular juvenile idiopathic arthritis (pJIA) in pediatric patients. This application relies on FDA's previous finding of safety and effectiveness of methotrexate for the reference product, Methotrexate Sodium Tablets NDA 008085 (Dava Pharmaceuticals, Inc).

The clinical pharmacology data submitted to support this application is from the relative bioavailability study SG02-03. Results of Study SG02-3 demonstrated that the proposed oral solution has comparable bioavailability to the reference product, methotrexate tablets. Additionally, a food effect study (SG02-04) showed that the presence of food reduces the Cmax of the proposed oral solution by 50% and delays the Tmax in 1.5 hour compared to the fasted state. The food effect finding is consistent with that of observed with the tablet formulation, as per reference product labeling. The reader is referred to the Clinical Pharmacology Review by Dr. Salaheldin Hamed from Division of Clinical Pharmacology V (dated April 15, 2016) for the review of these studies.

The approval of this NDA was withheld by deficiencies identified by the Agency during the inspection of the drug product manufacturing and packaging facility [redacted] (b) (4) in [redacted] (b) (4). A Complete Response Action Letter was issue on June 29, 2016. On October 25, 2016, Silvergate Pharmaceuticals, Inc. has submitted the current amendment to NDA 208400 in response to the Complete Response.

The Office of Clinical Pharmacology recommends approval of the proposed oral solution formulation provided that the Applicant and the Agency come to an agreement regarding the labeling language. The recommendations regarding the labeling sections pertaining to clinical pharmacology are provided in section 2 on this memorandum.

2. LABELING RECOMMENDATIONS

The review team conducted a thorough review of the label content to delete information that does not relate to the route of administration, dosage form and indications of this product. Label statements were also revised to be consistent with current labeling practice and remove scientifically outdate information, as appropriate. The published literature was utilized to provide scientific information relevant to the application and labeling.

The proposed changes to labeling statements pertaining to clinical pharmacology are listed below. Labeling statements to be removed are shown in ~~red strikethrough~~ font and suggested labeling to be included is shown in underline blue font.

7 DRUG INTERACTIONS

7.1 Effect of Other Drugs on XATMEP

(b) (4)

[redacted]

Reviewer's comment: *The language pertaining to nonsteroidal anti-inflammatory drugs (NSAID) interaction should be removed because it is not clinically relevant to the MTX dose level obtained with this formulation/dosage. While NSAIDs interactions with high-dose methotrexate therapy is well documented, the risk of methotrexate toxicity with low-dose MTX therapy and concurrent use of NSAIDs appears low based on published reviews.*[1,2,3]

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[1] Hall JJ, Bolina M, Chatterley T, Jamali F. Interaction between low-dose methotrexate and nonsteroidal anti-inflammatory drugs, penicillins, and proton pump inhibitors: a narrative review of the literature. Ann Pharmacother. 2016 Oct 3. pii: 1060028016672035. [Epub ahead of print]

[2] Bourré-Tessier J1, Haraoui B. Methotrexate drug interactions in the treatment of rheumatoid arthritis: a systematic review. J Rheumatol. 2010;37:1416-21.

[redacted] (b) (4) *Oral Antibiotics*

[redacted] (b) (4)

***Reviewer's comment:** This language was deleted because it is not clinically relevant.*

Penicillins may reduce the renal clearance of methotrexate; increased serum concentrations of methotrexate with concomitant hematologic and gastrointestinal toxicity have been observed with methotrexate. Monitor patients accordingly [*see Warnings and Precautions (5.1, 5.4)*]. [redacted] (b) (4)

Trimethoprim/sulfamethoxazole has been reported [redacted] (b) (4) to increase bone marrow suppression in patients receiving methotrexate [redacted] (b) (4) Monitor patients accordingly [*see Warnings and Precautions (5.1)*].

***Reviewer's comment:** MTX interactions with penicillins have been reported with low-dose methotrexate regimen, albeit associated with other patient factors such as renal impairment and oldage.[4,5,6] The concomitant use of MTX (including low dose between 5 and 15 mg per week) with trimethoprim/sulfamethoxazole was implicated as risk factor for developing bone marrow suppression in a retrospective case-control study[7] and in 17 case-reports.[2]*

[redacted] (b) (4) *Hepatotoxins*

The potential for increased hepatotoxicity when methotrexate is administered with other hepatotoxic agents has not been evaluated. However, hepatotoxicity has been reported in such cases. [redacted] (b) (4) Monitor patients receiving [redacted] (b) (4) XATMEP [redacted] (b) (4) with other potential hepatotoxins (e.g., azathioprine, retinoids, and sulfasalazine) [redacted] (b) (4) for possible [redacted] (b) (4) signs of hepatotoxicity.

[redacted] (b) (4) *Probenecid*

[redacted] (b) (4)

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[3] Colebatch AN, Marks JL, van der Heijde DM, Edwards CJ. Safety of nonsteroidal antiinflammatory drugs and/or paracetamol in people receiving methotrexate for inflammatory arthritis: a Cochrane systematic review. J Rheumatol. 2012;39(suppl 90):62-73

[4] Yamamoto K, Sawada Y, Matsushita Y, Moriwaki K, Bessho F, Iga T. Delayed elimination of methotrexate associated with piperacillin administration." Ann Pharmacother. 1997, 31: 1261-2.

[5] Mayall B, Poggi G, Parkin JD "Neutropenia due to low-dose methotrexate therapy for psoriasis and rheumatoid arthritis may be fatal." Med J Aust. 1991; 155: 480-4.

[6] Lim AYN, Gaffney K, Scott DGI. Methotrexate-induced pancytopenia: Serious and under-reported? Our experience of 25 cases in 5 years. Rheumatology. 2005;44:1051-55.

[7] Al-Awadhi A, Dale P, McKendry R. Pancytopenia associated with low dose methotrexate therapy: a regional survey. J Rheumatol, 1993;20:1121-25.

***Reviewer's comment**: This interaction language should be removed. MTX serum albumin binding is around 50%. Therefore, drug interactions mediated by plasma protein-biding displacement are unlikely to have clinical relevance.*[8]

(b) (4) [redacted] Probenecid may reduce renal elimination of methotrexate. Consider alternative drugs. (b) (4) [redacted]

***Reviewer's comment:** This interaction is mechanistically justified: probenecid is a well-documented inhibitor of the renal organic anion transporters (OAT) 1 and 3; while MTX is mainly eliminated renally, with renal tubular secretion mediated by OATs.*[9] *A case-report showed that MTX-probenecid interaction was clinically relevant at low dose MTX therapy (7.5 mg/week oral), albeit associated with other patients factors such as renal impairment and old age.*[10]

(b) (4) [redacted]

***Reviewer's comment:** This interaction language should be removed. The combined use of MTX with the conventional disease-modifying anti-rheumatic drugs hydroxychloroquine, and/or sulfasalazine has been evaluated in several controlled trials and is generally well-tolerated.* [11,12,13]

7.2 Effect of XATMEP on Other Drugs

(b) (4) [redacted] —*Theophylline*

Methotrexate may decrease the clearance of theophylline; monitor theophylline levels (b) (4) [redacted] when (b) (4) [redacted] with XATMEP.

***Reviewer's comment:** MTX interaction with theophylline, a narrow therapeutic index drug, was reported in a dedicated DDI study. After 6 weeks of treatment, the mean theophylline apparent clearance was 19% lower in the MTX treated-group (15 mg/week intra-muscular). Plasma levels of methotrexate were not measured in this study.*[14] *The interaction mechanism is unknown.*

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[8] Benet LZ, Hoener BA Changes in plasma protein binding have little clinical relevance. Clin Pharmacol Ther. 2002; 71: 115-21.

[9] Takeda M, Khamdang S, Narikawa S, et al. Characterization of methotrexate transport and its drug interactions with human organic anion transporters. J Pharmacol Exp Ther, 302: 666-71, 2002.

[10] Basin KS, Escalante A, Beardmore TD. Severe pancytopenia in a patient taking low dose methotrexate and probenecid." J Rheumatol. 1991;18: 609-10.

[11] Hazlewood GS, Barnabe C, Tomlinson G, Marshall D, Devoe DJ, Bombardier C. J Methotrexate monotherapy and methotrexate combination therapy with traditional and biologic disease modifying anti-rheumatic drugs for rheumatoid arthritis: A network meta-analysis. Cochrane Database Syst Rev. 2016; 8:CD010227.

[12] Clegg DO, Dietz F, Duffy J, et al. Safety and efficacy of hydroxychloroquine as maintenance therapy for rheumatoid arthritis after combination therapy with methotrexate and hydroxychloroquine. J Rheumatol. 1997;24:1896-902

[13] Saunders SA, Capell HA, Stirling A, Vallance R, Kincaid W, McMahon AD, Porter DR. Triple therapy in early active rheumatoid arthritis: a randomized, single blind, controlled trial comparing step-up and parallel treatment strategies. Arthritis Rheum. 2008; 58:1310-7.

[14] Glynn-Barnhart AM1, Erzurum SC, Leff JA, Martin RJ, Cochran JE, Cott GR, Szefler SJ. J Allergy Clin Immunol. 1991;88:180-6. Effect of low-dose methotrexate on the disposition of glucocorticoids and theophylline.

12 CLINICAL PHARMACOLOGY

12.2 Pharmacodynamics

Two reports describe *in vitro* methotrexate inhibition of DNA precursor uptake by stimulated mono-nuclear cells, and another describes in animal polyarthritis partial correction by methotrexate of spleen cell hyporesponsiveness and suppressed IL 2 production. Other laboratories, however, have been unable to demonstrate similar effects. [redacted] (b) (4) [redacted] (b) (4)

Reviewer's comment: *This statement is deleted because it is pertains to mechanism of action.*

[redacted] (b) (4)

Reviewer's comment: *This statement is deleted because it is not applicable to the proposed indications.*

12.3 Pharmacokinetics

Absorption

[redacted] (b) (4)

Reviewer's comment: *This language is deleted because it is not applicable to the proposed pediatric indications and the dosage/formulation.*

[redacted] (b) (4)

Reviewer's comment: *Per the Clinical Pharmacology Labeling Guidance, although bioequivalence or relative bioavailability may be a factor in the approvability of an application (e.g., 505(b)(2) applications), the term "bioequivalence" or the comparative PK data generally should not be included in the labeling.*

In [redacted] (b) (4) pediatric patients with ALL, oral absorption of methotrexate [redacted] (b) (4) appears to be dose dependent; the absorption of doses greater than 40 mg/m^2 is significantly less than that of lower doses. The extent of oral absorption [redacted] (b) (4) [redacted] (b) (4) ranges from 23% to 95% [redacted] (b) (4) [redacted] (b) (4) [redacted] (b) (4) [redacted] and the time to peak concentration (T_{max}) ranges from 0.[redacted]7 hours to 4 hours after an oral dose of 15 mg/m^2 [redacted]

[redacted] in [redacted] pediatric patients with pJIA, [redacted] mean serum concentrations were 0.59 micromolar (range, 0.03 to 1.40) at 1 hour, 0.44 micromolar (range, 0.01 to 1.00) at 2 hours, and 0.29 micromolar (range 0.06 to 0.58) at 3 hours[redacted], following oral administration of methotrexate at a dose of 6.4 mg/m2/week t[redacted] 11.2 mg/m2/week.

Effect of Food

The administration of XATMEP with food did not affect the area under the curve (AUC), but decreased the maximal concentrations (Cmax) by 50% and delayed the absorption.

Distribution

After intravenous administration, the initial volume of distribution is approximately 0.18 L/kg (18% of body weight) and steady-state volume of distribution is approximately 0.4 to 0.8 L/kg (40% to 80% of body weight).

Methotrexate competes with reduced folates for active transport across cell membranes by means of a single carrier-mediated active transport process. At serum concentrations greater than 100 micromolar, passive diffusion becomes a major pathway by which effective intracellular concentrations can be achieved.

Methotrexate in serum is approximately 50% protein bound. [redacted]

Methotrexate does not penetrate the blood-cerebrospinal fluid barrier in therapeutic amounts when given orally. [redacted]

[redacted] *Elimination*

[redacted] In adults, the half-life of methotrexate, following administration of low dose methotrexate (less than 30 mg/m^2), ranges from 3 hours to 10 hours. [redacted] (b) (4) (b) (4)

In pediatric patients receiving methotrexate for ALL [redacted] (b) (4) (6.3 mg/m2 to 30 mg/m2), [redacted] (b) (4) the terminal half-life has been reported to range from 0.7 to 5.8 hours [redacted] (b) (4) (b) (4)

In pediatric patients receiving methotrexate for JIA (3.75 mg/m2 to 26.2 mg/m2), the terminal half-life has been reported to range from 0.9 hours to 2.3 hours.

Metabolism

[redacted] (b) (4) M[redacted] (b) (4)ethotrexate undergoes hepatic and intracellular metabolism to polyglutamated f rms which can be converted back to methotrexate by hydrolase enzymes. These polyglutamates act as inhibitors of dihydrofolate reductase and thymidylate synthetase. Small amounts of methotrexate polyglutamates may remain in tissues for extended periods. The retention and prolonged drug action of these active metabolites vary among different cells, tissues and tumors. A small amount of metabolism to 7-hydroxymethotrexate may occur at doses commonly prescribed. [redacted] (b) (4) (b) (4) The aqueous solubility of 7-hydroxymethotrexate is 3 to 5 fold lower than the parent compound. Methotrexate is partially metabolized by intestinal flora after oral administration.

Excretion

Renal excretion is the primary route of elimination and is dependent upon dosage and route of administration. With IV administration, 80% to 90% of the administered dose is excreted unchanged in the urine within 24 hours. There is limited biliary excretion amounting to 10% or less of the administered dose. Enterohepatic recirculation of methotrexate has been proposed.

Renal excretion occurs by glomerular filtration and active tubular secretion. Nonlinear elimination due to saturation of renal tubular reabsorption has been observed in [redacted] (b) (4) patients at doses between 7.5 mg and 30 mg. Impaired renal function, as well as concurrent use of drugs such as weak organic acids that also undergo tubular secretion, can markedly increase methotrexate serum levels.

[redacted] (b) (4)

Methotrexate clearance [redacted] (b) (4) decreases[redacted] (b) (4)at higher doses. Delayed drug clearance has been identified as one of the major factors responsible for methotrexate toxicity. [redacted] (b) (4) (b) (4) (b) (4) When a patient has delayed drug elimination due to compromised renal function, a third space effusion, or other causes, methotrexate serum concentrations may remain elevated for prolonged periods.

(b) (4)

[redacted]

Reviewer's comment: *Statements pertaining to the pharmacokinetics of adult indications, dosage (high-dose) and routes of administration not approved for this product are deleted.*

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/s/

MANUELA GRIMSTEIN
03/31/2017

STACY S SHORD
03/31/2017
I concur

ANSHU MARATHE
03/31/2017

CLINICAL PHARMACOLOGY MEMORANDUM

Application Number:	208400, Original-2
Product:	XATMEP (Methotrexate Oral Solution)
Proposed Indication:	Polyarticular juvenile idiopathic arthritis (pJIA)
Applicant:	Silvergate Pharmaceuticals, Inc.
OCP Division:	Division of Clinical Pharmacology 2
Primary Reviewer:	Manuela Grimstein, PhD
Team Leader:	Anshu Marathe, PhD

A 505(b)(2) NDA application was submitted by Silvergate Pharmaceuticals, Inc. for Methotrexate Oral Solution, 2.5 mg/mL (XATMEP), on August 31, 2015. The application was administratively split into Original-1 (Division of Hematology Products [DHP]) and Original-2 (DPARP). The PDUFA goal date for NDA 208400 is June 30, 2016.

The applicant is seeking approval of this product for the treatment of pediatric patients with acute lymphoblastic leukemia (ALL), as part of a combination regimen, and the management of children with active polyarticular juvenile idiopathic arthritis (pJIA) who have had an insufficient therapeutic response to, or are intolerant of, an adequate trial of first line therapy, including full-dose nonsteroidal anti-inflammatory agents. The proposed weekly doses are 10 mg/m^2 and 20 mg/m^2 for the pJIA and ALL indications respectively.

This submission relies on the FDA's previous finding of safety and effectiveness of NDA 008085 (Methotrexate sodium oral tablets, Dava Pharmaceuticals). Two clinical pharmacology studies were submitted. A relative bioavailability study (SG02-03) comparing the oral solution formulation to the immediate release tablet formulation and a food effect study (SG02-04) where the proposed oral solution was administered with or without food. The reader is referred to the Clinical Pharmacology Review by Dr. Salaheldin Hamed from Division of Clinical Pharmacology V (DARRTs date 04/15/2016) for the review of these studies. Results showed that the proposed oral solution formulation and the immediate release tablets, at a 2.5 mg dose, have similar area under the curve (AUC) and maximum plasma concentration (CMAX), and that administration with food reduces the CMAX of the oral solution formulation by 50%.

Recommendation: The reviewer is in agreement with the findings in Dr. Hamed's review and recommends approval of NDA 208400 from a clinical pharmacology perspective

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/s/

MANUELA GRIMSTEIN
06/02/2016

ANSHU MARATHE
06/02/2016

CLINICAL PHARMACOLOGY REVIEW

NDA Number:	208400
Submission Type; Code:	505(b)(2)
Applicant Name:	Silvergate Pharmaceuticals, Inc.
Submission Dates:	August 31, 2015
Brand Name:	Xatmep
Generic Name	Methotrexate Sodium
Dosage Form:	Oral Solution
Dosage Strengths:	2.5 mg/mL
Proposed Indication:	Acute Lymphoblastic Leukemia (ALL) Polyarticular Juvenile Idiopathic Arthritis (pJIA)
OCP Division:	Division of Clinical Pharmacology V
Primary Reviewer:	Salaheldin Hamed, Ph.D.
Team Leader:	Bahru Habtemariam, Ph.D.

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1 EXECUTIVE SUMMARY

This application is a 505(b)(2) NDA submitted by Silvergate Pharmaceuticals, Inc. requesting the approval of Xatmep (methotrexate oral solution, 2.5 mg/mL) for the treatment of pediatric patients with acute lymphoblastic leukemia (ALL), as part of a combination regimen, and the management of children with active polyarticular juvenile idiopathic arthritis (pJIA) who have had an insufficient therapeutic response to, or are intolerant of, an adequate trial of first line therapy, including full-dose nonsteroidal anti-inflammatory agents.

To support this NDA, the applicant submitted results of a relative bioavailability study where the oral solution formulation was compared to the immediate release tablet formulation, the reference listed drug (RLD). The applicant also conducted a food effect study where the proposed oral solution was administered with or without food. Results from the two studies showed that the proposed oral solution formulation and the immediate release tablets, at a 2.5 mg dose, have similar area under the curve (AUC) and maximum plasma concentration (C_{MAX}) and that administration with food reduces the C_{MAX} of the oral solution formulation by 50% (**Table 1**).

Table 1: Summary of Relative BA and Food Effect Findings

Study	Parameter	Geometric Mean Ratio (90% CI)
BA study	AUC_{inf}	98 (95,102)
	C_{MAX}	107 (100, 115)
Food effect	AUC_{inf}	96 (91, 102)
	C_{MAX}	50 (47,54)

1.1 Recommendations

The Office of Clinical Pharmacology/Division of Clinical Pharmacology V has reviewed Silvergate's 505(b)(2) NDA requesting the approval of Xatmep, methotrexate sodium oral solution, and recommends the proposed product for approval from a clinical pharmacology perspective

Decision	Acceptable to OCP?			Comment
Overall	Yes ☒	No ☐	NA ☐	Approvable based on the established bioequivalence to the reference drug
Evidence of Effectiveness	Yes ☐	No ☐	NA ☒	
Proposed dose for general population	Yes ☒	No ☐	NA ☐	The proposed dosing of this product is based on the recommended dosing in the reference drug label
Proposed dose selection for	Yes ☐	No ☐	NA ☒	

others				
Pivotal BE	Yes ☒	No ☐	NA ☐	A pivotal BE study was conducted, the PK exposure was comparable between Xatmep (methotrexate oral solution) and the reference immediate release tablets
Labeling	Yes ☐	No ☒	NA ☐	Revised labeling is proposed

1.2 Phase 4 Commitments

None

1.3 Summary of Important Clinical Pharmacology and Biopharmaceutics Findings

The NDA for Xatmep (methotrexate sodium oral solution) was submitted under 505(b)(2) regulation referencing methotrexate sodium tablets developed by Dava (NDA 008085) as the RLD. The currently approved and marketed methotrexate products are for neoplastic, rheumatology, and psoriasis indications in adults and pediatrics. However, the applicant's only proposed indications are juvenile polyarticular rheumatoid arthritis (JRA) and acute lymphoblastic leukemia (ALL) in pediatric patients. The recommended doses for the JRA and ALL indications are 10 and 20 mg/m^2, respectively. Because the applicant is seeking approval for rheumatology as well as ALL indication, this NDA is being reviewed by the DPARP and DHP clinical divisions.

Below are summaries of the two clinical studies that are intended to support the proposed NDA:

- Study SG02-03 evaluated the PK of the proposed oral solution product in comparison to the reference product (2.5 mg immediate release tablets). The reported results indicate that methotrexate C_{MAX} and AUC of the proposed oral solution are similar to those of the immediate release tablet (See **Table 1**).
- Study SG02-04 evaluated the effect of food on the relative BA of the proposed methotrexate oral solution formulation. The reported results also indicate that food intake reduces C_{MAX} (see **Table 1**) and delays median T_{MAX} (range) from 0.83 h (0.5-1) in the fasted state to 2 h (0.66-4) in the fed state.

The relative BA/BE study was conducted at a lower dose of 2.5 mg whereas the recommended dose for the ALL indication is 20 mg/m^2. According to the methotrexate package insert, methotrexate exhibits saturable absorption at higher doses (>30 mg/m^2). The applicant provided a literature reference (PMID: 646750) demonstrating that intravenous methotrexate prepared as an oral solution has comparable exposure to that of methotrexate tablets up to doses of 25 mg/m^2. Therefore, the relative BA findings at 2.5 mg dose are deemed representative of the relative BA of methotrexate at the recommended dose (20 mg/m^2) for the ALL indication. Furthermore, any potential lack of bioequivalence at higher doses is not expected to have a meaningful therapeutic consequence since methotrexate treatment is titrated to achieve a therapeutic target in ALL patients.

The food effect finding, where food is shown to reduce the C_{MAX} of the oral solution formulation, is consistent with the effect of food on the tablet formulation, as per reference product labeling and the published literature (PMID: 15914465). The proposed labeling and the reference drug label do not provide specific recommendation regarding administration of the respective products with food. Since AUC and C_{trough} tend to be the relevant exposure parameters for efficacy, the reduced C_{MAX} is not expected to have any therapeutic consequence. However, for methotrexate exposure-response relationship for efficacy has not been shown for any exposure parameter (PMID: 24936744).

2 QUESTION BASED REVIEW

2.1 General Attributes of the Drug

2.1.1 What are the highlights of the chemistry and physical-chemical properties of the drug substance and the formulation of the drug product?

<u>Drug Substance:</u>

Methotrexate sodium ($C_{20}H_{22}N_8O_5$, MW = 454.45 g/mol) is a yellow to orange crystalline powder. The structure of methotrexate sodium (**Figure 1**) contains one asymmetric carbon atom, located in the glutamic acid side chain.

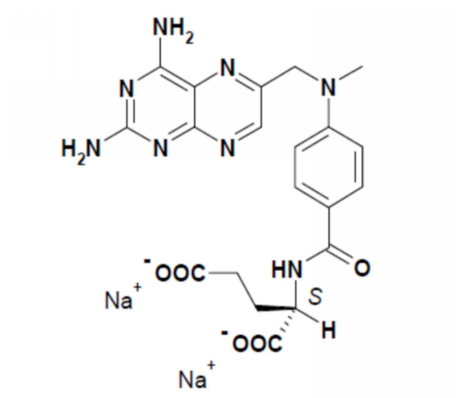

Figure 1: Structure of Methotrexate Sodium

Methotrexate is practically insoluble in water; however, methotrexate salt is freely soluble at higher pH (> 7).

<u>Drug Product:</u>

The proposed product is a ready-to-use aqueous solution containing 2.5 mg/mL of methotrexate.

2.1.2 What are the proposed mechanism of action and therapeutic indications?

<u>Mechanism of Action:</u>

- **Neoplastic Disease:** as currently labeled, methotrexate inhibits dihydrofolate reductase, which mediates the reduction of dihydrofolates to tetrahydrofolates before they can be utilized as carriers of one-carbon groups in the synthesis of purine nucleotides and thymidylate. Therefore, methotrexate interferes with DNA synthesis, repair, and replication. Actively proliferating cells such, as malignant cells, bone marrow cells, fetal cells, buccal and intestinal mucosa, and cells of the urinary bladder are generally more sensitive to the effect of methotrexate. When cellular proliferation in malignant tissues is greater than in most normal tissues, methotrexate may impair malignant growth without excessive damage to normal tissues.

- **Rheumatoid Arthritis:** as currently labeled, the mechanism of action in rheumatoid arthritis is unknown; it may affect immune function.

Proposed Indications:

- **Acute Lymphoblastic Leukemia (ALL):** Xatmep is indicated for the treatment of pediatric patients with acute lymphoblastic leukemia (ALL), as part of a combination regimen.
- **Rheumatoid Arthritis:** Xatmep is indicated for the management of children with active polyarticular juvenile idiopathic arthritis (pJIA) who have had insufficient therapeutic response to, or intolerant of, an adequate trial of first-line therapy, including full-dose nonsteroidal anti-inflammatory agents (NSAIDs).

2.1.3 What are the proposed dosages and routes of administration?

The proposed product is a solution for oral administration. In pediatric patients with acute lymphoblastic leukemia, the recommended dosing of Xatmep, in multi-agent combination chemotherapy maintenance regimens, is 20 mg/m^2 given once weekly.

2.2 General Clinical Pharmacology

2.2.1 What are the design features of the clinical pharmacology and the clinical studies used to support dosing or claims?

To support this 505(b)(2) NDA, the applicant conducted a relative bioavailability and a food effect study. Details of the two studies are provided in **Table 2**.

Table 2: Summary of Clinical Pharmacology Studies

Study (N)	Study Objective	Study Design	Treatments (Dose, Dosage Form, Route)	Subjects (M/F) Age: mean (Range)
SG02-03 (N=31)	To assess bioavailability of Xatmep, 2.5 mg/mL compared to an IR methotrexate tablet, 2.5 mg, under fasted conditions in healthy adult subjects.	Randomized, open-label, single-dose, two-period, two-treatment, two-way, crossover study.	A: Methotrexate oral solution,1 mL × 2.5 mg/mL (fasted) B: Methotrexate oral tablet, 1 × 2.5 mg (fasted)	34 healthy adults (34/0) Age: mean 34.41 years (19 to 54)

SG02-04 (N=21)	To assess the effect of food on the relative bioavailability of methotrexate oral solution.	Randomized, open-label, single-dose, two-period, two-treatment, two-way, crossover study.	A: Methotrexate oral solution, 1 mL × 2.5 mg/mL (fasted) B: Methotrexate oral solution, 1 mL × 2.5 mg/mL (fed)	24 healthy adults (24/0) Age: mean 36.92 years (18 to 55)

2.2.2 What are the PK characteristics of the drug?

The following PK information is obtained from the currently approved labels for oral and parenteral methotrexate and is relevant to the populations in which this product will be administered.

Absorption – In adults and pediatric leukemic patients, oral absorption appears to be dose dependent and varies widely at higher doses. Peak serum levels are reached within one to two hours. At doses of 30 mg/m^2 or less, methotrexate is generally well absorbed with a mean bioavailability of about 60%.

Distribution –After intravenous administration, the initial volume of distribution is approximately 0.18 L/kg (18% of body weight) and steady-state volume of distribution is approximately 0.4 to 0.8 L/kg (40% to 80% of body weight). Methotrexate competes with reduced folates for active transport across cell membranes by means of a single carrier-mediated active transport process. Methotrexate in serum is approximately 50% protein bound.

Metabolism – methotrexate undergoes hepatic and intracellular metabolism to polyglutamated forms which can be converted back to methotrexate by hydrolase enzymes. A small amount of metabolism to 7-hydroxymethotrexate may occur at doses commonly prescribed. Methotrexate is partially metabolized by intestinal flora after oral administration.

Half-Life – The terminal half-life of methotrexate is approximately 3 to 10 hours in patients receiving doses less than 30 mg/m^2. At higher doses, the terminal half-life is 8 to 15 hours. In pediatric patients receiving methotrexate for acute lymphocytic leukemia (6.3 to 30 mg/m^2), or for JIA (3.75 to 26.2 mg/m^2), the terminal half-life has been reported to range from 0.7 to 5.8 hours or 0.9 to 2.3 hours, respectively.

Excretion – Renal excretion, via glomerular filtration and active tubular secretion, is the primary route of elimination and is dependent upon dosage and route of administration. With IV administration, 80% to 90% of the administered dose is excreted unchanged in the urine within 24 hours. There is limited biliary excretion amounting to 10% or less of the administered dose. Nonlinear elimination due to saturation of renal tubular reabsorption has been observed in psoriatic patients at doses of 7.5 to 30 mg.

2.3 Intrinsic Factors

2.3.1 What intrinsic factors influence exposure and/or response, and what is the impact of any differences in exposure on efficacy or safety responses?

No formal studies were conducted to evaluate the effects of extrinsic factors on methotrexate exposure in this NDA.

2.4 Extrinsic Factors

2.4.1 What extrinsic factors influence exposure and/or response, and what is the impact of any differences in exposure on efficacy or safety responses?

No formal studies were conducted to evaluate the effects of extrinsic factors on methotrexate exposure in this NDA

2.5 General Biopharmaceutics

2.5.1 Based on the biopharmaceutics classification system principles, in what class is this drug and formulation? What solubility, permeability and dissolution data support this classification?

No information pertaining to the BCS classification was submitted in this NDA. However, methotrexate sodium exhibits high solubility in water. However, information available in the literature indicates that the free form exhibits low solubility and low permeability (i.e., BCS class IV).

2.5.2 What is the relative bioavailability of the proposed to-be-marketed formulation to the immediate release formulation?

Methotrexate oral solution was found to be equivalent, in terms of the AUC as well as C_{MAX}, to the immediate release oral tablet at the 2.5 mg dose.

Study SG02-03 was a randomized, open-label, two-period, two-treatment, crossover study in normal healthy male volunteers to determine the relative bioavailability of the proposed methotrexate oral solution versus methotrexate immediate release tablets under fasted conditions. The oral solution was found to be bioequivalent to the immediate release tablet (**Table 1** and **Figure 2**). A linear mixed effects model was used by this reviewer to analyze the fixed effects (formulation, period, and sequence) and the random effects (subjects nested within a sequence) on the PK parameters AUC and C_{MAX}. The geometric mean ratio of test (oral solution) to reference (tablets) along with the 90% confidence interval fell within 80% to 125%.

Out of the 34 subjects enrolled in the study, 31 subjects completed both periods (i.e. tablet and oral solution dosing). One subject was withdrawn due to an AE after being given a dose of the oral solution formulation. The AE was fungal infection that resolved after 6 days of treatment; the AE was considered mild by the investigator. Overall, methotrexate oral solution 2.5 mg/mL and methotrexate tablets 2.5 mg were well-tolerated in this study and the AE profiles were consistent with the known effects of the drug. No serious AE were reported (**Table 3**).

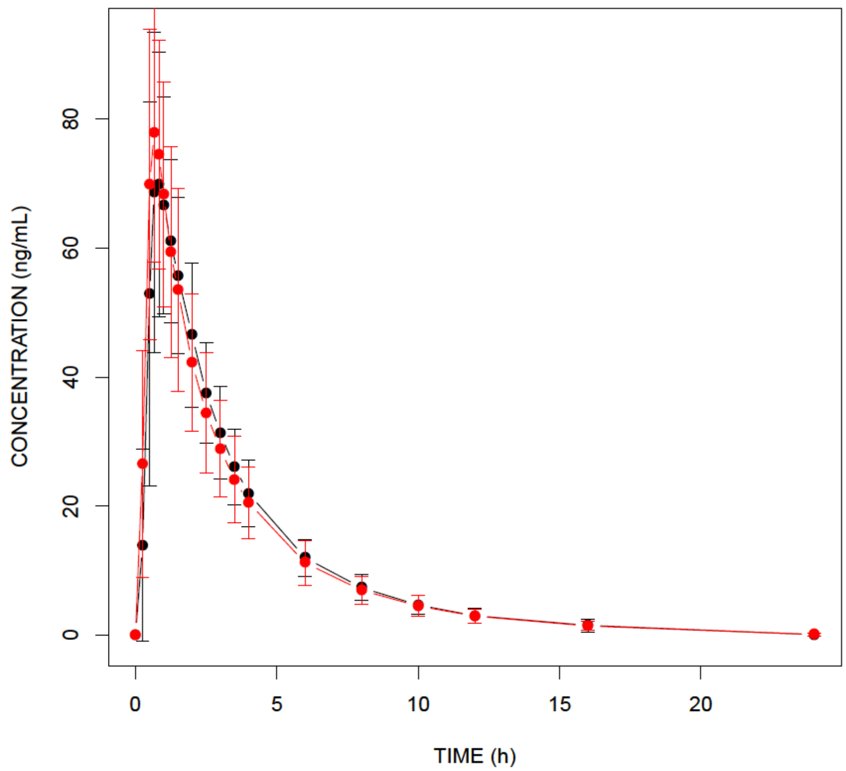

Figure 2: Mean PK profiles of immediate release tablets (black) and the proposed oral solution (red) in individual volunteers.

Table 3: Treatment-Emergent AE in Study SG02-03

System Organ Class	Preferred Term	Severity	Treatment A	Treatment B
Gastrointestinal disorders	Paraesthesia oral	Mild	0 (0.0%)	1 (3.1%)
Immune system disorders	Hypersensitivity	Mild	0 (0.0%)	1 (3.1%)
Infections and infestations	Fungal skin infection	Mild	1 (3.0%)	0 (0.0%)
Nervous system disorders	Headache	Mild	1 (3.0%)	1 (3.1%)
Respiratory, thoracic and	Nasal congestion	Mild	1 (3.0%)	0 (0.0%)
Skin and subcutaneous tissue disorders	Seborrhoeic dermatitis	Mild	1 (3.0%)	0 (0.0%)

Treatment A = methotrexate pediatric oral solution, 2.5 mg/mL (n=33)
Treatment B = methotrexate tablets, 2.5 mg (n=32)

2.5.3 What is the effect of food on the bioavailability of the drug from the dosage form?

Study SG02-04 was a randomized, open-label, single-dose, two-period, two-treatment, two-way, crossover relative bioavailability study of methotrexate oral solution under fasted and fed conditions in healthy adult male volunteers. Study volunteers were given a 2.5 mg dose of Xatmep after a 10-hour overnight fast (reference) or after consuming a standard high-calorie, high-fat breakfast meal in another period (test).

The AUC under fasted and fed conditions was essentially comparable; however, C_{MAX} was decreased by ~50% and T_{MAX} was delayed (median of 2 hours versus 0.83 hours) in the fed state.

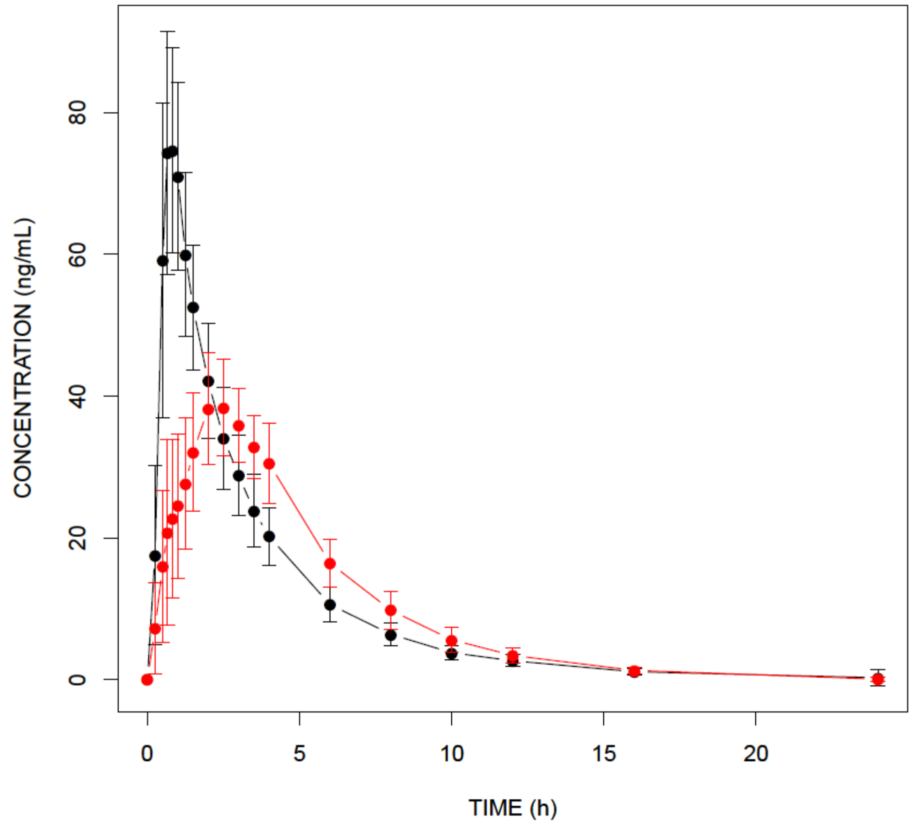

Figure 3: Mean PK profiles of Xatmep under fasted (black line) and fed (red line) conditions.

2.6 Analytical Section

2.6.1 How are the active moieties identified and measured in the plasma?

A method was initially employed for measuring the parent drug, methotrexate, and the metabolite, 7-hydroxymethotrexate. The parent drug and metabolite were extracted from 0.100 mL of plasma samples by solid phase extraction and high-performance liquid chromatography tandem mass spectrometry (LC-MS/MS). The method did not meet the accuracy and precision criteria for the metabolite, 7-hydroxymethotrexate. The method was validated for measuring only the parent drug, methotrexate (**Table 4**).

2.6.2 Which metabolites have selected for analysis and why?

The metabolite, 7-hydroxymethotrexate, was not measured due to assay validation issues. The metabolite is present at very low concentration – less than 10% of the total dose – at this low dose.

2.6.3 For all the moieties measured, is free, bound, or total measured?

The total concentration of methotrexate was measured in the two submitted studies.

2.6.4 What bioanalytical methods are used to assess concentrations?

The concentration of methotrexate was measured in plasma using a high performance liquid chromatography tandem mass spectrometry (HPLC-MS/MS) assay.

2.6.4.1 What is the range of the standard curve? How does it relate to the requirements for clinical studies? What is curve fitting technique?

The standard curve spanned a range of concentrations from 0.5 ng/mL to 500 ng/mL. The concentration range covered the observed C_{MAX} (up to120 ng/mL) and the lowest observed concentration (0.6 ng/mL at 16 hours). Weighted ($1/x^2$) linear regression was used for back calculation of methotrexate.

2.6.4.2 What are the lower and upper limits of quantitation?

The LLOQ was 0.5 ng/mL and the ULOQ was 500 ng/mL.

2.6.4.3 What are the accuracy, precision, and selectivity at these limits?

The within-run and between-run precision did not exceed 8% and accuracy did not exceed 7%.

2.6.4.4 What is the sample stability under conditions used in the study?

The stability of QC samples were assessed at room temperature (25 hours), freeze thaw cycles (5 cycles), and at conditions of use. The bias of all the stability samples was within the 15% acceptance criteria.

2.6.4.5 What is the QC sample plan?

QC samples ($n \geq 2$) at low (1.5 ng/mL), medium (25 ng/mL), and high (400 ng/mL) concentrations were included in each run. The accuracy and precision of the QC samples were well below the 15% limit for the accepted runs.

Table 4: Summary Methotrexate LC-MS/MS Method Validation

<table>
<tr><th>Validation Parameters</th><th colspan="6">Results</th></tr>
<tr><td>Short term stability studies</td><td colspan="6">Plasma: stable up to 25 hours at room temperature
Extract: stable up to 73 hours at room temperature</td></tr>
<tr><td>Reinjection reproducibility</td><td colspan="6">Reinjection reproducibility is acceptable</td></tr>
<tr><td rowspan="3">Recovery</td><td>Analyte</td><td>From</td><td>To</td><td>Internal</td><td>From</td><td>To</td></tr>
<tr><td>MTX (Run 2)</td><td>81%</td><td>86%</td><td>MTX-D$_3$ (Run 2)</td><td>80%</td><td>89%</td></tr>
<tr><td>MTX (Run 4)</td><td>76%</td><td>90%</td><td>MTX-D$_3$ (Run 4)</td><td>76%</td><td>82%</td></tr>
<tr><td rowspan="3">Matrix factor</td><td>Analyte</td><td>From</td><td>To</td><td>Internal standard</td><td>From</td><td>To</td></tr>
<tr><td>MTX (Run 2)</td><td>1.27</td><td>1.46</td><td>MTX-D$_3$ (Run 2)</td><td>1.26</td><td>1.50</td></tr>
<tr><td>MTX (Run 4)</td><td>1.11</td><td>2.01</td><td>MTX-D$_3$ (Run 4)</td><td>1.13</td><td>1.87</td></tr>
<tr><td rowspan="3">Assay selectively</td><td colspan="2" rowspan="2">Analyte</td><td colspan="4">Bias</td></tr>
<tr><td colspan="2">From</td><td colspan="2">To</td></tr>
<tr><td colspan="2">Methotrexate</td><td colspan="2">-19.0%</td><td colspan="2">12.6%</td></tr>
</table>

Detection specificity	Results meet acceptance criteria		
Long-term stability	QC samples are stable up to 67 days at -20°C		
Stock solution stability	Analytical method storage		
	Temperature	Time	Difference
	Target concentration 20.0 µg/mL		
	4°C	39 days	-0.83%
	Room temperature	26 hours	2.43%
	Target concentration 200 µg/mL		
	4°C	42 days	-3.45%
	Room temperature	21 hours	0.97%
Carryover	No significant carryover		

This is a representation of an electronic record that was signed electronically and this page is the manifestation of the electronic signature.

/s/

SALAHELDIN S HAMED
04/14/2016

BAHRU A HABTEMARIAM
04/15/2016

CLINICAL PHARMACOLOGY FILING FORM

Application Information

NDA/BLA Number	208400	**SDN**	0000
Applicant	Silvergate Pharmaceuticals. Inc.	**Submission Date**	31 August 2015
Generic Name	Methotrexate	**Brand Name**	TRADENAME
Drug Class	folate analog metabolic inhibitor		
Indication	(1) Treatment of pediatric patients with acute lymphoblastic leukemia (ALL) and (2) Management of [redacted] with active polyarticular juvenile idiopathic arthritis (pJIA).		
Dosage Regimen	(1) The recommended dose of TRADENAME, in multi-agent combination chemotherapy maintenance regimens, i [redacted] mg/m² given [redacted] weekly. (2) The recommended starting dose of TRADENAME for pJIA is 10 mg/m² given once weekly.		
Dosage Form	Solution	**Route of Administration**	Oral
OCP Division	DCP2	**OND Division**	DPARP

OCP Review Team	**Primary Reviewer(s)**	**Secondary Reviewer/ Team Leader**
Division	Manuela Grimstein, Ph.D.	Ping Ji, Ph.D.
Pharmacometrics		
Genomics		

Review Classification	☑ Standard ☐ Priority ☐ Expedited		
Filing Date	10/30/2015	**74-Day Letter Date**	11/13/2015
Review Due Date	6/2/2016	**PDUFA Goal Date**	6/30/2016

Application Fileability

Is the Clinical Pharmacology section of the application fileable?

☑ Yes

☐ No

If no, list reason(s)

Are there any potential review issues/ comments to be forwarded to the Applicant in the 74-day letter?

☐ Yes

☑ No

If yes, list comment(s)

Is there a need for clinical trial(s) inspection?

☐ Yes

☐ No

If yes, explain: Refer to the Clinical Pharmacology Filing Form of Dr. Salaheldin Hamed.

Clinical Pharmacology Package

Tabular Listing of All Human Studies	☑ Yes ☐ No	Clinical Pharmacology Summary	☑ Yes ☐ No
Bioanalytical and Analytical Methods	☑ Yes ☐ No	Labeling	☑ Yes ☐ No

Clinical Pharmacology Studies

Study Type	**Count**	**Comment(s)**
In Vitro Studies		
☐ Metabolism Characterization		

☐ Transporter Characterization			
☐ Distribution			
☐ Drug-Drug Interaction			
In Vivo Studies			
Biopharmaceutics			
☐ Absolute Bioavailability			
☑ Relative Bioavailability		1	SG02-03 (Module 5.3.1.2)
☐ Bioequivalence			
☑ Food Effect		1	SG02-04 (Module 5.3.1.2)
☐ Other			
Human Pharmacokinetics			
Healthy Subjects	☑ Single Dose	2	SG02-03 (Module 5.3.1.2) SG02-03 (Module 5.3.1.2)
	☐ Multiple Dose		
Patients	☐ Single Dose		
	☐ Multiple Dose		
☐ Mass Balance Study			
☐ Other (e.g. dose proportionality)			
Intrinsic Factors			
☐ Race			
☐ Sex			
☐ Geriatrics			
☐ Pediatrics			
☐ Hepatic Impairment			
☐ Renal Impairment			
☐ Genetics			
Extrinsic Factors			
☐ Effects on Primary Drug			
☐ Effects of Primary Drug			
Pharmacodynamics			
☐ Healthy Subjects			
☐ Patients			
Pharmacokinetics/Pharmacodynamics			
☐ Healthy Subjects			
☐ Patients			
☐ QT			
Pharmacometrics			
☐ Population Pharmacokinetics			
☐ Exposure-Efficacy			
☐ Exposure-Safety			

Total Number of Studies	**2**	**In Vitro**		**In Vivo**	2

Total Number of Studies to be Reviewed 2				2

Criteria for Refusal to File (RTF)		
RTF Parameter	**Assessment**	**Comments**
1. Did the applicant submit bioequivalence data comparing to-be-marketed product(s) and those used in the pivotal clinical trials?	☐Yes ☐No ☑N/A	The clinical/registration batch size is (b)(4) and the commercial batch size is (b)(4)
2. Did the applicant provide metabolism and drug-drug interaction information? (Note: RTF only if there is complete lack of information)	☐Yes ☐No ☑N/A	Cross-reference to NDA 008085
3. Did the applicant submit pharmacokinetic studies to characterize the drug product, or submit a waiver request?	☐Yes ☐No ☑N/A	Cross-reference to NDA 008085
4. Did the applicant submit comparative bioavailability data between proposed drug product and reference product for a 505(b)(2) application?	☑Yes ☐No ☐N/A	
5. Did the applicant submit data to allow the evaluation of the validity of the analytical assay for the moieties of interest?	☑Yes ☐No ☐N/A	
6. Did the applicant submit study reports/rationale to support dose/dosing interval and dose adjustment?	☐Yes ☐No ☑N/A	
7. Does the submission contain PK and PD analysis datasets and PK and PD parameter datasets for each primary study that supports items 1 to 6 above (in .xpt format if data are submitted electronically)?	☑Yes ☐No ☐N/A	
8. Did the applicant submit the module 2 summaries (e.g. summary-clin-pharm, summary-biopharm, pharmkin-written-summary)?	☑Yes ☐No ☐N/A	Pharmkin-written-summary is cross-referenced to NDA 008085 and findings from public literature.
9. Is the clinical pharmacology and biopharmaceutics section of the submission legible, organized, indexed and paginated in a manner to allow substantive review to begin? If provided as an electronic submission, is the electronic submission searchable, does it have appropriate hyperlinks and do the hyperlinks work leading to appropriate sections, reports, and appendices?	☑Yes ☐No ☐N/A	
Complete Application **10.** Did the applicant submit studies including study reports, analysis datasets, source code, input files and key analysis output, or justification for not conducting studies, as agreed to at the pre-NDA or pre-BLA meeting? If the answer is 'No', has the sponsor submitted a justification that was previously agreed to before the NDA submission?	☑Yes ☐No ☐N/A	

Criteria for Assessing Quality of an NDA (Preliminary Assessment of Quality) Checklist		
Data		
1. Are the data sets, as requested during pre-submission discussions, submitted in the appropriate format (e.g., CDISC)?	☑Yes ☐No ☐N/A	
2. If applicable, are the pharmacogenomic data sets submitted in the appropriate format?	☐Yes ☐No ☑N/A	
Studies and Analysis		
3. Is the appropriate pharmacokinetic information submitted?	☑Yes ☐No ☐N/A	
4. Has the applicant made an appropriate attempt to determine reasonable dose individualization strategies for this product (i.e., appropriately designed and analyzed dose-ranging or pivotal studies)?	☐Yes ☐No ☑N/A	
5. Are the appropriate exposure-response (for desired and undesired effects) analyses conducted and submitted as described in the Exposure-Response guidance?	☐Yes ☐No ☑N/A	
6. Is there an adequate attempt by the applicant to use exposure-response relationships in order to assess the need for dose adjustments for intrinsic/extrinsic factors that might affect the pharmacokinetic or pharmacodynamics?	☐Yes ☐No ☑N/A	
7. Are the pediatric exclusivity studies adequately designed to demonstrate effectiveness, if the drug is indeed effective?	☐Yes ☐No ☑N/A	Cross-reference to NDA 008085 for previous findings of safety and effectiveness in pediatrics. Not required. Methotrexate oral solution was granted Orphan Drug Designation on May 28 and August 28, 2015 for the treatment of ALL in pediatrics, and the management of oligoarticular JIA and pJIA in pediatrics, respectively.
General		
8. Are the clinical pharmacology and biopharmaceutics studies of appropriate design and breadth of investigation to meet basic requirements for approvability of this product?	☑Yes ☐No ☐N/A	
9. Was the translation (of study reports or other study information) from another language needed and provided in this submission?	☐Yes ☐No ☑N/A	

Filing Memo	
IS THE CLINICAL PHARMACOLOGY SECTION OF THE APPLICATION FILEABLE?	☑Yes ☐No
If the NDA/BLA is not fileable from the clinical pharmacology perspective, state the reasons and provide comments to be sent to the Applicant. Please identify and list any potential review issues to be forwarded to the Applicant for the 74-day letter.	
Comments to Sponsor:	None identified.

Manuela d L. T. Grimstein — 22 October, 2015

Reviewer Clinical Pharmacology — Date

Ping Ji — 25 October, 2015

Acting Team Leader — Date

This is a representation of an electronic record that was signed electronically and this page is the manifestation of the electronic signature.

/s/

MANUELA GRIMSTEIN
10/27/2015

PING JI
10/27/2015